

SMART INDIA HACKATHON 2026

Integrated Precision Guidance & Environmental Sensing Architecture

A safety-bounded project proposal for high-dynamics navigation, resilient sensing, and post-event environmental monitoring

Prepared for the Yantra India Limited precision-guidance-kit problem context

Conceptual academic research proposal · Version 1.0 · September 2026

Safety boundary

This proposal is limited to inert laboratory demonstrators, simulation, environmental sensing, and controlled shutdown. It excludes explosive design, target-specific engagement logic, lethal-effect optimization, and operational deployment guidance.

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1 Executive summary

This proposal describes an integrated architecture for a rapidly rotating, high-dynamics vehicle that transitions into a comparatively stable sensing platform and, after a controlled impact surrogate, into a benign environmental-monitoring node. The concept is not presented as a validated weapon or as a build-ready operational system. It is presented as a research program whose success is measured by falsifiable experiments, credible system interfaces, graceful failure behavior, and a strict safety boundary.

The central engineering problem is a moving reference frame. A spin-stabilized carrier rotates rapidly, while cameras, navigation antennas, computers, and actuators perform best when their reference frame is stable. The proposal responds architecturally rather than merely computationally: it separates a forward guidance section from the spinning carrier, damps relative rotation, and uses the same relative motion to study power recovery. It then layers event-driven perception, cooperative state estimation, conformal sensing surfaces, and safe post-event environmental observation on top of that stable reference frame.

The proposal in one sentence

Separate the sensing reference frame, recover useful energy from the motion that remains, fuse asynchronous observations conservatively, and end with a safe environmental-sensing demonstration rather than a harmful function.

1.1 What is known, what is proposed, and what must be proven

The research note combines established principles with ambitious system-level hypotheses. Keeping those categories separate makes the proposal more credible.

Evidence level	Meaning in this proposal	Examples
Established principle	Supported by textbook, industry, or adjacent-domain evidence.	Faraday-Lenz magnetic damping; event-camera asynchronous sensing; inertial navigation; risk-managed AI.
Engineering hypothesis	Plausible integration claim that needs simulation, coupon testing, or hardware-in-the-loop evidence.	Despin transient; harvested-power balance; conformal multi-band aperture; shock-isolated sensor carrier.
Competition target	A measurable result proposed for SIH, not a promise of operational performance.	Stable low-spin reference in a rig; bounded event classification; graceful network degradation.

1.2 Why the project matters

The project targets a difficult but general class of engineering systems: platforms that experience shock, rotation, vibration, limited energy, uncertain sensing, and short decision windows. The same design questions appear in disaster-response devices, infrastructure inspection, post-impact geophysical monitoring, and autonomous systems that must continue operating after a disruptive

event. The safe demonstrator therefore has dual-use value without requiring a harmful payload or engagement function.

1.3 Decision requested from a research evaluator

The recommended decision is to fund or mentor the **evidence-building phase**: system simulation, scaled rotating-interface experiments, event-stream data collection, mesh-network fault injection, inert shock-survival testing, and hardware-in-the-loop safety verification. Full-scale or flight-representative testing is explicitly outside this proposal's competition-stage scope and would require separate authority, facilities, legal review, and safety certification.

2 Project definition and design thesis

2.1 The problem

A rapidly rotating body creates four coupled difficulties. First, the sensor reference frame moves continuously. Second, high acceleration and vibration constrain mechanisms and electronics. Third, energy is limited while communication and perception create peak power demand. Fourth, a single failed sensor or node can corrupt the system if uncertainty is not represented explicitly.

A conventional response adds compensation. This proposal asks whether the constraint can be reduced before compensation is applied. The forward section is treated as an isolated avionics and sensing platform. The spinning carrier retains its stabilizing role. After a controlled inert impact surrogate, the platform changes mission phase and becomes a low-power environmental sensor.

2.2 Design thesis

The architecture has two transitions:

1. **Reference-frame transition:** a locked nose leaves the launch-like state, relative rotation is damped, and the guidance sensors operate from a more stable frame.
2. **Mission-purpose transition:** after an inert shock event, a protected sensor carrier deploys a sensing head, monitors bounded environmental categories, transmits status, and shuts down predictably.

Important limitation

The supplied research contains numerical claims that have not been independently validated. This proposal keeps them as preliminary design hypotheses and assigns each a test method. No number below should be interpreted as a certified operational specification.

2.3 Seven-pillar architecture

No.	Pillar	Plain-language role
1	Despun Nose Architecture	Creates a comparatively stable platform for sensing and control.
2	Liquid Mass Transfer Steering	Moves internal mass to create a slow, coarse trim effect; external surfaces provide faster control.

No.	Pillar	Plain-language role
3	Event-Driven Terminal Seeker	Reports changes rather than full frames, reducing redundant data and latency under difficult motion.
4	Spin-Harvesting Axial-Flux Generator	Converts relative rotation into electrical energy for buffering and low-power functions.
5	Cooperative Swarm Mesh Network	Allows several safe nodes to share time, position, confidence, and environmental estimates.
6	Metamaterial Conformal Multi-Function Aperture	Embeds communication and sensing functions into the structure rather than adding many protruding antennas.
7	Impact-Survivable Deployable Environmental Sensor	Protects inert electronics through a shock surrogate and observes bounded environmental events after deployment.

2.4 System dependency logic

P1: stable reference frame

P4: energy buffer

P3 + P5 + P6: perception, networking, aperture

P7: safe post-event monitoring

The critical path is therefore **P1 → P4 → {P3, P5, P6}**, with P7 providing the safe dual-use demonstration. If P1 does not achieve the intended despin behavior, the proposed fallback is to disable the advanced functions and retain only a conservative inert simulation or coarse, non-operational control demonstration.

3 Real-time concept of operations

This section explains what the system would do in time order. The words “launch,” “flight,” and “impact” refer to simulation or approved inert laboratory surrogates in this proposal.

3.1 Phase 0 — pre-test initialization

Before an experiment begins, the controller verifies the article identity, sensor health, time synchronization, battery state, mechanical lock state, and safe-mode availability. The test supervisor selects a bounded scenario. The system records a baseline so that later changes can be distinguished from pre-existing vibration, temperature, or radio noise.

No downstream function is permitted to become active from software alone. Demonstrator interlocks should require independent physical conditions, a test fixture state, and an operator-controlled enable. For an SIH prototype, those conditions can be represented by a keyed enclosure, fixture sensor, and software state machine without reproducing hazardous launch hardware.

3.2 Phase 1 — locked high-dynamics surrogate

In the high-acceleration portion of a rig or digital simulation, the nose and carrier remain mechanically coupled. Canards or equivalent external control surfaces are stowed. Internal fluid is constrained. The purpose is to prevent relative motion from damaging interfaces before the safe transition point. The real-time estimator reads inertial sensors at high rate, but it does not attempt to run the full perception stack. Instead, it checks whether measured acceleration and angular rate remain inside the test envelope. If they do not, the controller records a fault and remains in a safe state.

3.3 Phase 2 — controlled release and despin study

At the transition event, the test fixture or simulation releases the nose. Relative rotation is measured by two independent signals: inertial correlation and an optical or magnetic tachometer. A contactless brake model predicts the damping response. The controller does not need to “command” the passive brake; it observes the response and decides whether the platform has entered the stable-reference envelope.

The research note correctly identifies an important feasibility point: eddy-current braking alone produces a torque that falls with relative speed, so a pure brake may take too long. The proposal therefore treats hybrid damping as a hypothesis to test rather than a guaranteed result. The rig should compare passive magnetic damping, aerodynamic-equivalent drag, bearing damping, and their combinations.

3.4 Phase 3 — navigation and control handover

When residual spin, vibration, and estimator uncertainty fall below thresholds, the system enables the next layer. Navigation combines inertial measurements with a simulated satellite or laboratory reference. The event-driven camera is not continuously used. It wakes when a trigger indicates that a visual change contains useful information, such as a contrast boundary or motion feature.

The control hierarchy is intentionally slow-to-fast. Internal mass transfer provides coarse trim over a longer interval. External canards, if represented in the laboratory demonstrator, provide faster differential control. A supervisor limits rate, authority, and duty cycle. If the estimator loses confidence, the system freezes the last safe command and returns to a bounded fallback model.

3.5 Phase 4 — cooperative estimation

A safe node or simulator broadcasts timestamped estimates rather than raw sensor streams. Each message includes state, uncertainty, sensor provenance, and a sequence number. A receiving node rejects stale, contradictory, or low-confidence messages. This is the difference between “networked” and “trustworthy networked” behavior.

The network is tested with packet loss, clock bias, delayed messages, and node dropout. The expected result is not perfect cooperation. The expected result is **graceful degradation**: an individual node continues with local sensing, and the system can quantify how much accuracy or confidence was lost.

3.6 Phase 5 — inert impact surrogate

The test article enters a controlled shock fixture. A sacrificial shell and crushable zone lengthen the impact event, reducing the acceleration transmitted to the protected sensor carrier. No energetic payload, target selector, or harmful action exists in any state.

Immediately after the surrogate event, the carrier performs a self-check. It measures shock history, checks enclosure continuity, validates sensor calibration, and decides whether deployment is safe. If deployment is not safe, the device remains sealed and records the failure.

3.7 Phase 6 — environmental monitoring and shutdown

After deployment, the sensor head establishes a quiet baseline. A low-power threshold detector watches for changes in vibration, sound, magnetic field, temperature, or passive infrared signal. A possible event wakes the processor, which collects a short observation window, extracts bounded features, assigns a confidence score, records the result, and sends a non-sensitive status packet.

The event classifier is deliberately restricted to laboratory categories such as wind, rainfall, controlled-rig vibration, footsteps from test equipment, and animal-like disturbances. It does not identify people for engagement, select targets, or initiate harmful action. At the end of the predefined window, or immediately after a fault, the controller commits the log and enters safe shutdown.

Real-time state	What the system measures	What it does next
LOCKED	Acceleration, angular rate, fixture state	Remain inert; reject premature transitions.
TRANSITION	Relative rate, bearing response, shock	Estimate despin; permit or deny handover.
GUIDANCE-SIM	Navigation, confidence, control demand	Fuse sources; apply bounded commands in simulation.
MESH	Packets, timestamps, uncertainty	Fuse trustworthy messages; continue locally if partitioned.
IMPACT-SURVIVE	Shock history, enclosure, sensor health	Deploy only if safe; otherwise record failure.
MONITOR	Threshold events and environmental features	Wake, classify bounded categories, transmit, log.
SAFE-OFF	Fault, timeout, battery, enclosure state	Stop actuation; preserve data; close the test.

4 Feature-by-feature explanation

4.1 Pillar 1 — Despun Nose Architecture

Easy explanation. Imagine a small instrument platform mounted inside a rotating outer body. During the high-dynamics segment, the platform is locked. After the safe transition, it is allowed to rotate relative to the body and is damped toward a low-spin state. The platform can then keep its camera, inertial sensors, and antenna in a more useful orientation.

Research basis. The proposed mechanism combines a launch-lock concept, bearings, and contactless magnetic damping. Eddy currents arise when a conductor experiences changing magnetic flux; by Lenz's law, the induced magnetic field opposes the change and can produce smooth, non-contact damping [1]. The source research itself finds that a pure eddy-current brake would likely be too slow for the proposed transition, so hybrid damping must be experimentally measured rather than assumed.

What must be demonstrated. A scaled rotating rig should measure angular-rate decay, temperature, vibration, alignment, and failure behavior. The acceptance criterion is not a claimed full-scale RPM number; it is repeatable evidence that a designed interface can reduce relative motion while maintaining sensor integrity and a documented fallback.

4.2 Pillar 2 — Liquid Mass Transfer Steering

Easy explanation. Moving liquid inside a sealed container shifts the center of mass. In free flight, gravity does not create a useful internal torque because the whole vehicle accelerates together. The useful effect comes from the changed mass distribution and the aerodynamic response that follows. This correction is a major strength of the research. The original 200 g concept was too small for meaningful authority. The proposal therefore labels 1.0–1.2 kg as a preliminary research configuration, not a final specification. A laboratory demonstrator can replace the liquid with a safe surrogate and measure center-of-mass shift, pressure response, thermal behavior, and repeatability.

Why combine it with external surfaces? Internal mass transfer is slow and does not protrude into the flow. External surfaces are faster but experience drag, heating, and structural loads. A combined architecture is therefore conceptually complementary: slow trim inside, fast control in the simulation or non-flight rig.

4.3 Pillar 3 — Event-Driven Terminal Seeker

Easy explanation. A conventional camera repeatedly sends complete pictures, including pixels that did not change. An event camera reports local brightness changes asynchronously. ESA describes this as high temporal resolution and high dynamic range with less redundant information, making event-based sensing relevant to navigation and landing research [2].

Real-time behavior. The sensor produces timestamped events. A motion estimator accumulates them over short windows. A confidence gate decides whether the observation is strong enough to influence the state estimate. If not, the system continues on inertial prediction and reports lower confidence. This “do nothing when uncertain” behavior is more important than a high average classification score.

4.4 Pillar 4 — Spin-Harvesting Axial-Flux Generator

Easy explanation. The same relative motion that is being damped can also move magnetic flux through coils. The generator converts part of that motion into electrical energy, which is buffered in a capacitor or battery.

Key real-time question. Net energy matters, not peak generation. The experiment must measure generated power, brake torque, electronics demand, storage losses, and temperature at the same time. The hypothesis is accepted only if the energy recovered is useful without creating unacceptable mechanical or thermal penalties.

4.5 Pillar 5 — Cooperative Swarm Mesh Network

Easy explanation. Several safe nodes can compare notes. Each node shares an estimate and how certain it is. A node does not blindly trust a message; it checks time, sequence, origin, consistency, and uncertainty.

Validation. A multi-node simulator should inject packet loss, delayed messages, clock offsets, biased sensors, and node failures. The primary result is graceful degradation compared with isolated nodes. A network that becomes confidently wrong is a failure, even if it appears highly connected.

4.6 Pillar 6 — Metamaterial Conformal Multi-Function Aperture

Easy explanation. Instead of adding separate antennas and sensors that occupy volume and disturb the surface, the proposal studies a layered structural skin that performs multiple electromagnetic functions.

This is the least mature pillar. The proposal treats it as a simulation and coupon-manufacturing study. It must address mutual coupling, temperature drift, manufacturing tolerance, impact survivability, and electromagnetic compatibility. A conventional internal antenna remains the fallback. No competition claim should depend on the conformal aperture succeeding on the first prototype.

4.7 Pillar 7 — Impact-Survivable Deployable Environmental Sensor

Easy explanation. A protected electronics carrier rides through a controlled impact surrogate, opens a compact sensor head if its self-check passes, monitors the environment for a fixed time, communicates status, and shuts down.

The safe research function is environmental observation. Candidate sensors include a geophone, microphone, magnetometer, passive temperature or infrared sensor, and laboratory-safe ranging. The software classifies bounded categories and records uncertainty. The system cannot initiate a harmful action because no such function is present in the architecture.

5 Research methodology and evidence plan

5.1 Work packages

WP	Work package	Evidence produced	Priority
1	Architecture and safety case	CAD cross-section, interface register, state machine, hazard boundary, fallback map	Essential
2	Dynamics and control simulation	6-DOF model, sensitivity sweeps, uncertainty bands, trajectory comparisons	Essential
3	Rotating-interface experiment	Angular-rate decay, damping coefficient, temperature, alignment, repeatability	High
4	Event perception	Synthetic and recorded event streams, latency, confidence calibration, false-positive analysis	High
5	Cooperative mesh	Packet-loss and bias-injection results; local-vs-network comparison	High

WP	Work package	Evidence produced	Priority
6	Shock-survival demonstrator	Instrumented acceleration, carrier continuity, sensor function after test	Essential
7	Deployment and monitoring	Repeated deployment, baseline stabilization, bounded event logs, shutdown integrity	High
8	Hardware-in-the-loop integration	Mode transitions, fault response, interlock tests, reproducible demo script	Essential

5.2 Measurement philosophy

Every claimed result should include a measured quantity, an uncertainty or repeatability statement, a test fixture description, and a known limitation. For example, “despin works” is not sufficient. A strong result is “the scaled interface reduced relative angular rate by a measured factor across repeated trials, with temperature and alignment remaining within the fixture’s envelope.”

5.3 Digital model and Monte Carlo analysis

The digital model should separate truth, sensor models, estimator, controller, and safety supervisor. Monte Carlo runs should vary mass properties, damping, sensor bias, timing, packet loss, temperature, and single failures. The output should be a distribution of error and confidence, not a single optimistic trace.

The source research recommends comparing unguided, canard-only, mass-transfer-only, and combined cases. That comparison is useful as a systems-engineering experiment. It should be reported as a simulation result with assumptions, not as a field-performance promise.

6 Feasibility, risks, and mitigations

6.1 Feasibility verdict

Component	Indicative maturity	Confidence	Primary risk to test
Despun interface and magnetic damping	Medium	Medium	Launch-like shock, bearing behavior, air-gap sensitivity, hybrid damping response
Canard-like control in a non-flight rig	Adjacent-domain basis	Medium-high	Clearance, actuation under load, control authority assumptions
Internal mass-transfer surrogate	Early	Medium-low	Small center-of-mass shift, fluid temperature, mass budget
Event-driven perception	Adjacent-domain basis	Medium	Motion blur is reduced, but confidence can still fail under clutter

Component	Indicative maturity	Confidence	Primary risk to test
Energy harvesting	Early	Medium	Net power after damping and storage losses
Cooperative mesh	Adjacent-domain basis	Medium-high	Timing, Doppler-like effects in simulation, biased nodes
Conformal aperture	Early	Low-medium	Repeatability, coupling, thermal drift, impact damage
Inert environmental sensor	Medium	Medium-high	Shock attenuation, deployment, calibration, shutdown

6.2 Risk register

Risk	Why it matters	Mitigation and go/no-go evidence
Reference frame does not stabilize	Advanced sensing and networking lose value.	Measure residual rate independently; retain a conservative fallback and do not claim precision performance.
Mass-transfer authority is too small	The internal mechanism may not justify its complexity.	Use corrected free-flight physics; compare against a safe surrogate and a canard-like simulation.
Energy harvest is net-negative	Braking may cost more than generation provides.	Measure torque, power, storage loss, and temperature together.
Perception confidence is poor	False positives can create unsafe decisions.	Bound the labels, expose confidence, gate actions, and use NIST-style risk management [3].
Network makes estimates worse	Correlated or biased messages can amplify error.	Inject bias and packet loss; reject stale or inconsistent messages.
Impact damages electronics	The post-event monitoring objective fails.	Instrument the surrogate, iterate crushable layers, and accept “sealed/no deploy” as a valid safe outcome.
Safety boundary is violated	The project becomes unacceptable for an academic demonstration.	Inert articles, independent interlocks, no harmful state, predictable shutdown, documented supervision.

6.3 Safety and responsible-innovation case

The safety case is a design feature, not a footnote. The proposal uses a positive safety boundary: only inert articles and controlled fixtures; only bounded environmental categories; no energetic payload; no target selection; no engagement logic; no operational deployment instructions; and a shutdown state that preserves data and removes actuation authority.

For the AI component, the project should document intended use, prohibited use, data provenance, confidence calibration, error analysis, human oversight, and fault response. NIST describes its AI Risk Management Framework as a voluntary framework for incorporating trustworthiness considerations into the design, development, use, and evaluation of AI systems [3]. That framing is appropriate for the event classifier and cooperative estimator.

7 Implementation roadmap

7.1 SIH-stage plan

Stage	Milestone	Demonstration	Exit evidence
1	Architecture baseline	CAD, interface map, safety state machine, dependency graph	All interfaces named and bounded
2	Digital dynamics	Repeatable simulation with uncertainty and single-failure cases	Assumptions and sensitivity recorded
3	Rotating rig	Safe scaled despin and generator experiment	Measured damping and thermal traces
4	Perception and mesh	Event-stream classifier plus multi-node fault-injection demo	Latency, confidence, degradation curves
5	Shock and deployment	Instrumented inert impact surrogate and sensor-head mock-up	Survival and shutdown evidence
6	Integrated HIL demo	Operator-supervised end-to-end mode sequence	No unsafe state reachable

7.2 Post-SIH validation pathway

A credible next phase would proceed from coupon-level and spin-table tests to instrumented shock testing, RF chamber validation, and only then any flight-representative subscale experiment. Each step requires a separate facility, legal review, test plan, and independent safety approval. The proposal does not authorize or prescribe those activities; it identifies them as future validation gates.

8 Judge-facing value proposition

8.1 What is genuinely novel

The novelty is not that every component is new. The novelty is the system-level interaction among reference-frame management, energy recovery, event-driven perception, cooperative estimation,

multifunctional structures, and safe post-event sensing. The architecture also makes its critical path visible: if the reference frame does not stabilize, the downstream gains are not silently assumed.

8.2 Why the proposal is credible

The research note contains a useful correction: the original 200 g mass-transfer calculation overstated free-flight authority. Correcting it improves the proposal because it shows that the team distinguishes a mathematically valid expression from a physically relevant model. The same discipline should be applied to every headline number.

Recommended pitch

An integrated, self-monitoring architecture that removes a difficult reference-frame constraint, measures the energy and uncertainty costs of its own decisions, and ends with safe environmental observation after a controlled event.

8.3 Questions evaluators may ask

Question	Evidence-based response
"Is this just a collection of unrelated subsystems?"	No. The dependency graph makes the coupling explicit: reference-frame stability enables sensing; relative motion enables both damping and energy recovery; power and structure constrain networking and aperture design; the post-event sensor provides the safe integrated demonstration.
"What is the weakest link?"	The despin interface is the critical path. That is why it receives the earliest rig test and a defined fallback. A strong proposal identifies its weakest link before a reviewer does.
"Why not use only external control surfaces?"	External surfaces are fast but incur drag, heating, and structural loads. Internal mass transfer is slow and trim-oriented. The proposal tests whether their roles are complementary rather than assuming that they are.
"What happens when the network fails?"	Nodes fall back to local sensing, stale data are rejected, and confidence decreases. The system is evaluated on graceful degradation, not connectivity alone.
"What happens after the impact surrogate?"	A self-check decides whether deployment is safe. The sensor observes bounded environmental categories, transmits status, and shuts down at timeout or fault. No harmful function exists.

9 Conclusion

This project is best understood as a staged systems-engineering research program. Its strongest contribution is a coherent way to connect mechanics, control, sensing, power, networking, structure, and safe post-event monitoring while admitting where the concept is immature. The competition-

stage deliverable should therefore emphasize interfaces and evidence: a transparent digital model, a small number of compelling safe experiments, clear uncertainty, graceful failure, and a non-negotiable safety boundary.

The proposal is promising enough to justify SIH-level research and prototyping. It is not yet a validated operational system. The path from concept to credible result is practical: prove the rotating interface, correct the control physics, quantify net power, constrain perception, test network degradation, instrument shock survival, and demonstrate predictable shutdown.

10 References

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11 Appendix — compact glossary

Term	Plain-language meaning
Axial-flux generator	A generator in which magnetic flux crosses the machine roughly along its axis; here it is studied as a relative-motion energy harvester.
Confidence gating	A rule that allows an observation to influence the estimate only when its uncertainty is sufficiently low.
Despin	Reduction of relative rotational speed between a spinning carrier and an internal platform.
Event camera	A vision sensor that reports local brightness changes asynchronously instead of sending only complete image frames.
Graceful degradation	A controlled reduction in capability after a fault, rather than an uncontrolled or unsafe failure.
Hardware-in-the-loop	A test in which real hardware interacts with a real-time simulation.
Lenz's law	The induced magnetic effect opposes the change that produced it.
Mass-transfer steering	Changing internal mass distribution to shift the center of mass and alter the aerodynamic response.
Metamaterial aperture	A structured surface designed to shape electromagnetic behavior; here treated as an early research hypothesis.
Reference frame	The coordinate system in which motion and sensor measurements are interpreted.
Safe shutdown	A final state that stops actuation, preserves records, and prevents further operation after timeout or fault.
State estimator	Software that combines sensor measurements and a motion model to infer position, velocity, orientation, and uncertainty.
SWaP-C	Size, weight, power, and cost—the practical budget that often determines feasibility.

Final note

This document is a proposal for research and demonstration. It should be reviewed by qualified safety, legal, mechanical, electrical, software, and test professionals before any physical experiment.